

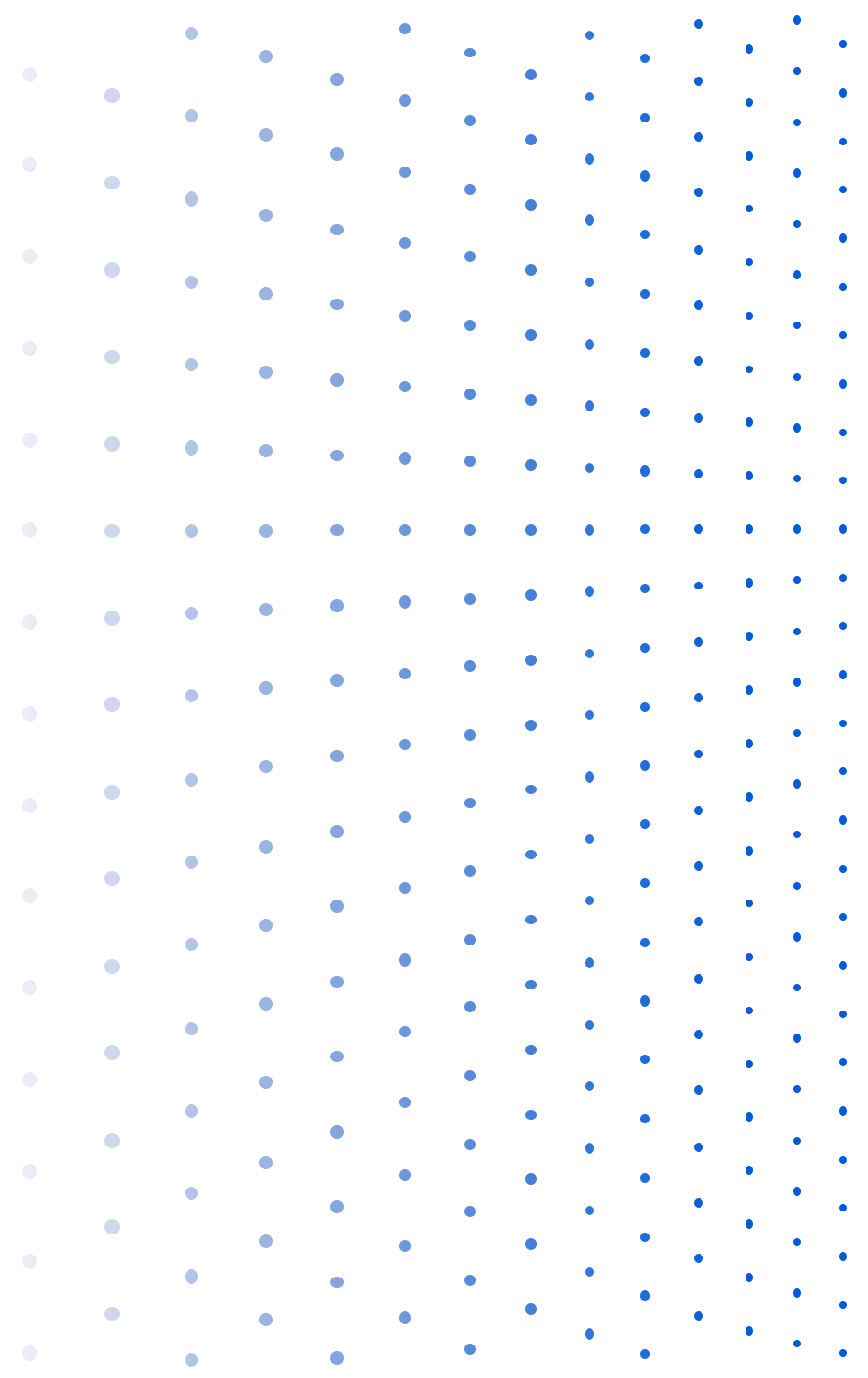


ProcDNA Case Study: Nexavir | PrimeRx Flow

Alert-to-Treatment & Commercial Opportunity Analysis

2026-27

This document is solely for the use of client personnel. No part of it may be circulated, quoted or reproduced for distribution outside of the client organization without prior written approval of ProcDNA.



PrimeRx Flow – Business & Case Study Overview

Business Overview:

- **PrimeRx** is a pharmaceutical company operating across the U.S. healthcare landscape, and its flagship product is **Nexavir** (DRG001), a treatment designed for high clinical need and broad prescriber adoption.
- You have been assigned to analyze PrimeRx's Rx flow, HCP, account, sales, and alert datasets collected over the last 18 months and provide recommendations to help the sales team act on the most valuable commercial opportunities.

Case Study Overview:

This case study explores a pharma commercial analytics problem: **doctors (HCPs) are diagnosing patients who may need treatment, but not all of them are prescribing our drug.**

You will use clinical alert, doctor affiliation, and prescription sales data to identify:

- HCPs and accounts with high prescription and alert activity
- Doctors receiving strong clinical signals but writing few or no prescriptions
- Accounts with low prescription activity relative to their doctor base
- HCPs and accounts that should be prioritized by the sales team

The goal is to connect **diagnostic activity** with **prescribing behavior** and translate the insights into a clear, data-driven action plan

PrimeRx Flow – Datasets Overview

You are provided with three datasets capturing different aspects of the patient treatment journey:

- **Alerts:** Clinical signals indicating potential treatment need at the doctor level
- **Sales:** Prescription activity across doctors and therapies
- **Affiliation:** Mapping of doctors to hospitals/healthcare accounts

These datasets can be linked using **HCP_ID**. Your task is to connect clinical activity with prescribing behavior to identify missed opportunities and recommend actionable strategies for the commercial team.

Dataset	Rows	Key Columns	Dataset Purpose
Alerts	9,537	Alert_ID (unique alert), Alert_Date, HCP_ID, Lab_Result (positive/negative)	Clinical alerts tied to doctors indicating potential treatment need
Sales	5,917	Prescription_ID (unique Rx event), Prescription_Date, HCP_ID, Drug_Name, Drug_ID, Prescription_Volume (units/scripts prescribed)	Prescription activity across doctors and drugs
Affiliation	216	HCP_ID, Account_ID, Account_Name	Maps doctors to hospitals/clinics/accounts

Question 1: Who are the most active doctors and are they helping our product?

Your starting universe is only the doctors who appear in the **Alerts** dataset i.e., doctors for whom at least one alert has been generated. Do not include any doctor who has no alert record.

From this universe, produce two separate Top 10 lists using two different ranking metrics:

- **Ranking 1 - Top 10 by Alert Volume:** Which 10 doctors have the highest total number of alerts generated?
- **Ranking 2 - Top 10 by Prescription Volume:** Which 10 doctors **from the alerts universe** have written the highest prescription volume in the Sales data?

Ranking 1 - Top 10 by alert volume

Rank	HCP_ID	Total Alerts
1	HCP041	200
2	HCP011	198
3	HCP019	190
4	HCP020	188
5	HCP032	188
6	HCP021	185
7	HCP014	178
8	HCP036	176
9	HCP022	175
10	HCP007	173

Ranking 2 - Top 10 by prescription volume

Rank	HCP_ID	Total Prescription Volume
1	HCP185	2900
2	HCP062	2380
3	HCP019	2277
4	HCP163	2233
5	HCP126	2196
6	HCP211	2196
7	HCP154	2187
8	HCP215	2155
9	HCP088	2084
10	HCP004	2081

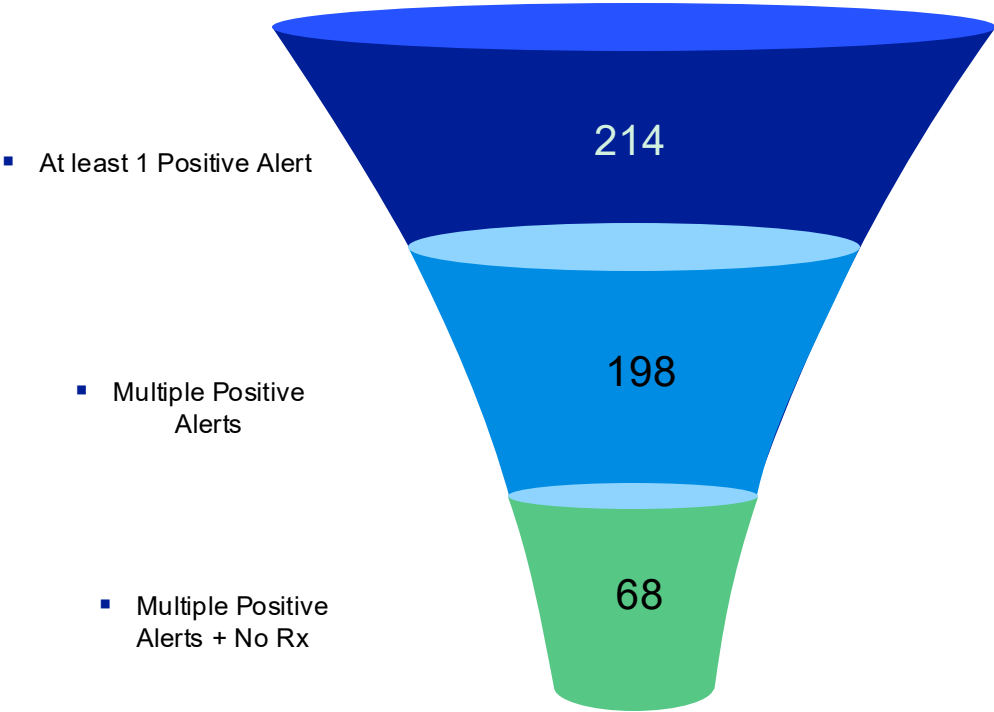
Question 2: Which doctors see the need but don't act on it?

Identify all doctors from the alerts universe who received **positive lab test alerts**, indicating that the patient likely required treatment. Evaluate whether these doctors showed any prescription activity for **any drug within 30 days** following their **first positive alert**. Among doctors with no prescribing activity during this post-alert window, rank the Top 10 HCPs by total **positive alert volume**.

Additionally, create a funnel showing HCPs with at least one positive alert, multiple positive alerts, and multiple positive alerts but zero prescribing activity within 30 days.

Positive-alert doctors and 30-day prescription follow-up

Rank	HCP_ID	Positive Alerts
1	HCP041	127
2	HCP011	123
3	HCP021	116
4	HCP042	101
5	HCP023	99
6	HCP007	96
7	HCP026	78
8	HCP009	67
9	HCP001	63
10	HCP002	59



Question 3: Among our top doctors, how much business is going to competitors?

Use the **Top 10 doctors by total alert volume** from Question 1.
For these 10 doctors, calculate the share of total prescription volume contributed by:

- **Our Drug (DRG001)**
- **Competitor Drugs (all other drugs)**

Next, identify which competitor drug accounts for the highest prescription volume and is capturing the largest share among your highest-alerting doctors.

Prescription mix for Top 10 alert-volume HCPs

Alert Rank	HCP_ID	Total Prescription Volume	% Our Drug Share	% Competitor Share
1	HCP041	0	0	0
2	HCP011	0	0	0
3	HCP019	2277	59.11	40.89
4	HCP020	570	25.44	74.56
5	HCP032	2075	59.13	40.87
6	HCP021	0	0	0
7	HCP014	944	71.08	28.92
8	HCP036	1895	64.75	35.25
9	HCP022	1363	62	38
10	HCP007	0	0	0

Most prescribed competitor drug among these HCPs

Competitor Drug Name	Drug_ID	Total Prescription Volume
Rivalex	DRG002	1470

Observations / Inferences

4 of the top 10 alert HCPs (**HCP041, HCP011, HCP021, HCP007**) have zero prescriptions for any drug; not even competitors. So, this isn't really a competitive loss, these doctors aren't prescribing at all yet. That is a different problem, needs other approach. For the other 6, competitor share ranges from **28.92% to 74.56%**. **Rivalex (DRG002)** is taking the biggest chunk of business away from us in this group.

Question 4: Which hospitals are sitting on the biggest untapped opportunity?

Using the Affiliation dataset, identify the **Top 10 accounts with the highest alert volume** based on activity generated by their affiliated doctors.
For these Top 10 accounts, calculate:

- Total Alert Count
- Our Drug Prescription Count (DRG001)
- Conversion Ratio - defined as the proportion of prescription count relative to alert count

Rank the accounts by lowest conversion ratio to identify high-alert accounts with weak prescription conversion.

Top 10 accounts by alert count

Rank	Account_ID	Total Alerts	Our Drug Prescription Count	Conversion Ratio
1	ACC001	3296	385	0.117
2	ACC003	1195	159	0.133
3	ACC002	2350	373	0.159
4	ACC004	530	164	0.309
5	ACC005	585	275	0.470
6	ACC006	493	298	0.604
7	ACC012	85	57	0.671
8	ACC009	100	77	0.770
9	ACC007	114	209	1.833
10	ACC010	92	295	3.207

Question 5: Which accounts represent the largest untapped prescription opportunity?

Using the Affiliation dataset, determine the **total number of doctors** associated with each account. Using prescription activity for Our Drug, calculate:

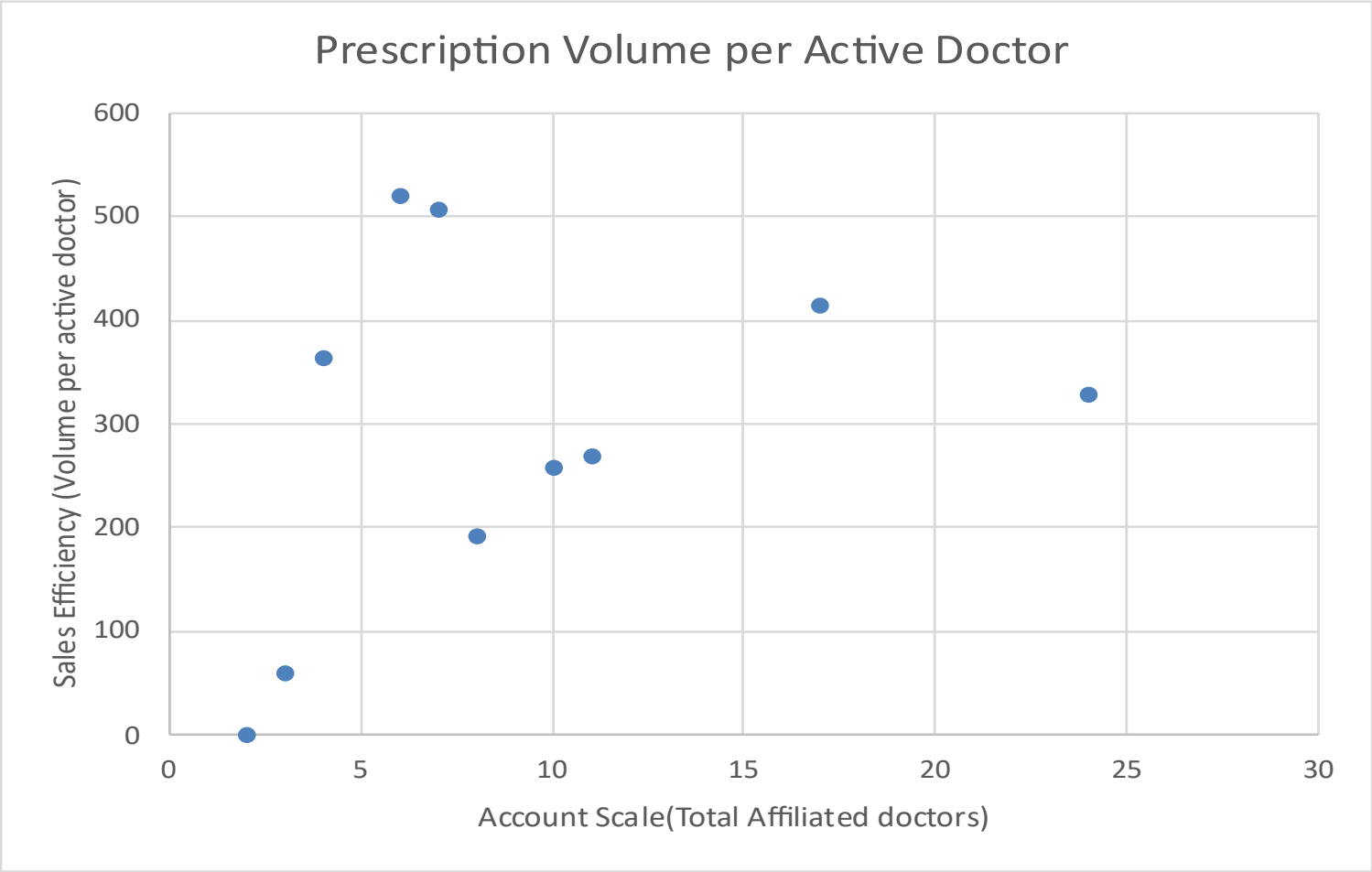
- The number of **active prescribers** within each account
- Total **prescription volume** at the account level

Finally, rank accounts by the **lowest prescriptions-per-active-doctor ratio** and identify the Top 10 accounts. Additionally, create a scatter plot comparing account size vs prescriptions-per-active-doctor ratio, highlighting potential underperforming accounts.

Account size vs company-drug sales efficiency

Rank	Account_ID	# Affiliated Doctors	# Active Doctors	Our Drug Prescription Volume	Prescription Volume per Active Doctor
1	ACC022	2	0	0	0
2	ACC017	3	1	59	59
3	ACC015	8	7	1347	192.429
4	ACC012	10	7	1804	257.714
5	ACC009	11	10	2699	269.9
6	ACC003	24	15	4947	329.8
7	ACC018	4	3	1095	365
8	ACC004	17	13	5402	415.538
9	ACC008	7	6	3049	508.167
10	ACC014	6	6	3126	521

Untapped Hospitals: Big Doctor Pools, Low Brand Adoption



What strategic inferences can be derived?

The biggest lever is not getting new prescribers, it's getting more volume out of doctors that are already active.

ACC003 (Riverside) is the easiest case: 24 affiliated doctors, but only 329.8 prescriptions per active doctor, well below ACC014's 521 despite Riverside having 4x the doctor base.

It is probably the competitor winning that gap, not a lack of engaged doctors.

Fastest path to growth are the mid-size, mostly active, but still mid-pack on volume accounts such as **ACC009** and **ACC012** because the doctor relationships are in place; it is a prescribing-behavior problem, not an access problem.

Question 6: Did alerts actually change doctor behavior - before vs. after?

Part A - Basic lift: Define a 90-day before and 90-day after window around each doctor's first positive alert. Calculate prescription volume for our drug in each window. Rank Top 10 HCPs by lift % with highest lift being Rank 1?

Part B - Segment the lift: Split all doctors into 3 behavioral bucket as mentioned in the formula table. Does the lift look different across the three groups? Which group shows strongest commercial impact?

Part A: HCP-level before / after lift

Rank	HCP_ID	Total Prescription Volume Before Alert	Total Prescription Volume After Alert	Lift %
1	HCP040	8	143	1687.5
2	HCP050	10	107	970
3	HCP006	32	249	678.125
4	HCP108	31	238	667.7419
5	HCP016	50	279	458
6	HCP159	42	232	452.381
7	HCP157	72	387	437.5
8	HCP211	42	223	430.9524
9	HCP209	64	335	423.4375
10	HCP125	49	214	336.7347

Part B: Segment Summary

Segment	HCPs	Avg Lift
New starters	64	N/A
Growers	25	171.44
Non-responders	125	-56.86
Total	214	

Segment showing Strongest Commercial Impact	New starters
---	--------------

Formula

Lift% : (Post Prescription Volume – Pre Prescription Volume) / (Pre Prescription Volume) * 100

- New starters: pre=0, post>0
- Growers: pre>0 and lift% ≥ +20%
- Non-responders: Everything else

Question 7: Commercial HCP Prioritization

A pharma company assign sales representatives to promote the drug to HCP. Your company's commercial team is planning the **next quarter's** outreach strategy for **DRG001**. Following are the field force assumptions:

- Each sales rep can complete **~8 effective HCP calls per day**
- Each rep works **~20 field days per month**
- The organization has **40 sales reps nationally**

Leadership wants to use this limited **field force capacity** efficiently by prioritizing the **HCP segments** identified in **Question 6** (New Starters, Growers, and Non-Responders).

Using the segment-level insights from Question 6, recommend how the company should allocate total rep call capacity across these segments. In your response:

- Estimate how much **outreach focus** each segment should receive
- Explain the reasoning behind your **allocation approach**
- Discuss the tradeoff between **growth opportunity, conversion likelihood, and sales effort** required across segments

Total rep call capacity per quarter
19200

Segment	HCPs	Suggested % of Rep Calls	Estimated Quarterly Calls	Estimated Rep Allocation
New starters	64	50	9600	20
Growers	25	30	5760	12
Non-responders	125	20	3840	8
Total	214	100	19200	40

Executive Summary

Core issue: Alerting identifies clinically relevant HCPs, but this does not lead to Nexavir prescriptions. There is a clear disconnect between the signal and prescription behavior.

- There is nearly no overlap between the top alerting and top prescribing physicians, with only 1 doctor in common. This indicates a weak connection between signal and action.
- 68 doctors, who were alerted multiple times, show no prescriptions in the next 30 days. This is the largest group experiencing conversion leakage.
- Four of the top alerting doctors are completely inactive, with no Nexavir or competitor prescriptions. This suggests low engagement rather than brand switching.
- Share loss mainly affects Rivalex (DRG002), which is taking most of the diverted business.
- High-potential accounts like Northside General (11.7%) and Eastview Medical (15.9%) show strong alert activity but weak conversion.
- Riverside Health System has the greatest potential, with approximately 2,392 possible prescriptions among inactive HCPs.
- Response varies by group: New Starters and Growers show strong improvement, while Non-Responders decline after the alert. This suggests diminishing returns or fatigue.

Recommended action: Focus rep efforts based on responsiveness: about 50% for New Starters, 30% for Growers, and 20% for Non-Responders. Prioritize field engagement with Northside, Eastview, and Riverside first.

Limitations: 18-month window may mask the seasonal effects. Drug-switching dynamics outside this dataset aren't captured.



Thank You!

This document is solely for the use of client personnel. No part of it may be circulated, quoted or reproduced for distribution outside of the client organization without prior written approval of ProcDNA.

LinkedIn www.procdna.com

South San Francisco | San Diego | Boston | New Jersey | Princeton | Chicago | Delhi | Pune

